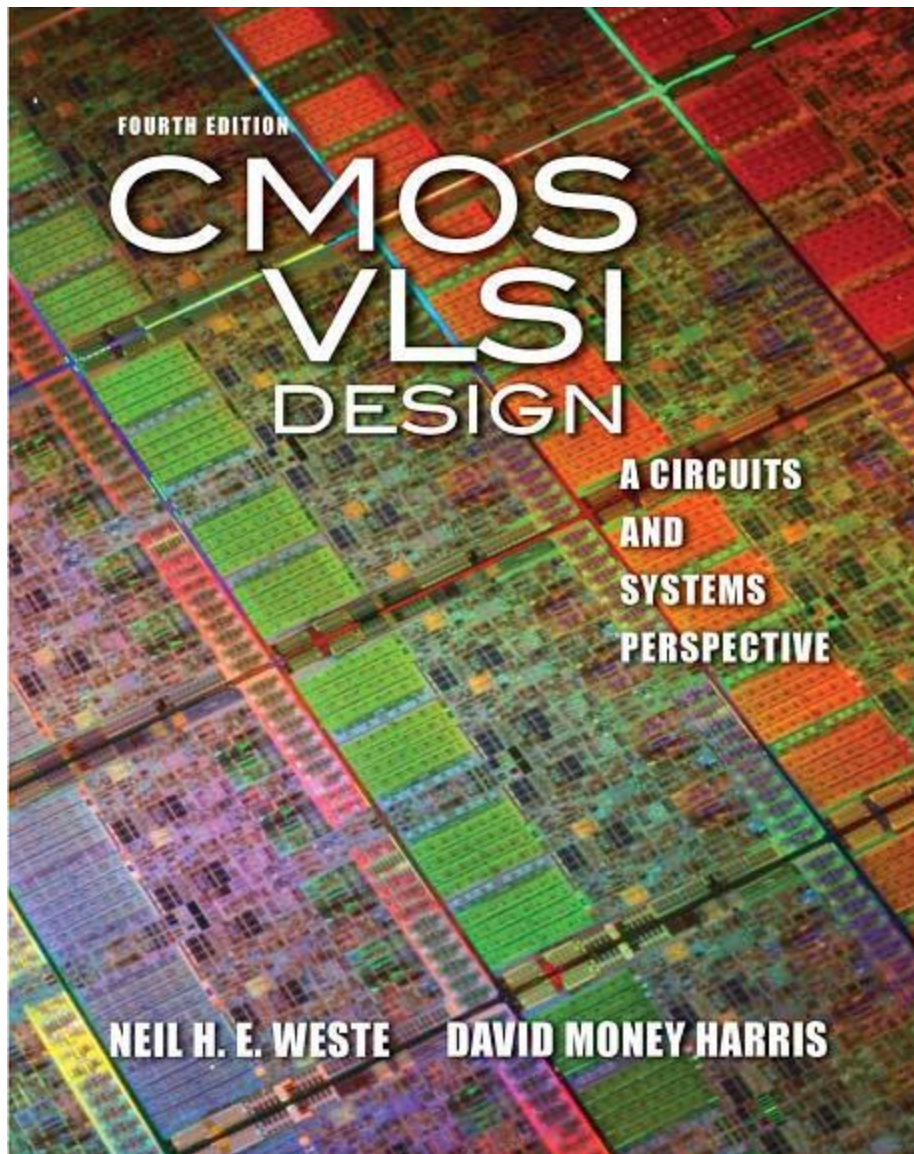




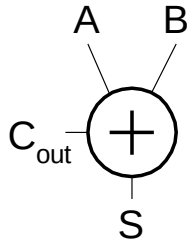
Chapter 11: Datapath

Lecture Content

- ❑ Single-Bit Addition
- ❑ Full Adder Design I (Static CMOS)
- ❑ Full Adder Design II (Mirror Adder)
- ❑ Full Adder Design III (Transmission Gate Full Adder)
- ❑ Full Adder Design IV (CPL)
- ❑ Full Adder Design V (Dual-rail domino)

Single-Bit Addition

Half Adder



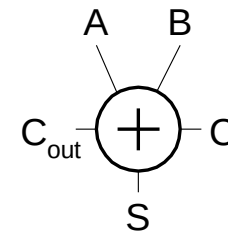
A	B	C_{out}	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$$S = A \oplus B$$

$$C_{out} = A \cdot B$$

Full Adder

A	B	C	C_{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



$$S = A \oplus B \oplus C$$

$$C_{out} = MAJ(A, B, C)$$

$$\begin{aligned}
 S &= \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC \\
 &= (A \oplus B) \oplus C = P \oplus C \\
 C_{out} &= AB + AC + BC \\
 &= AB + C(A + B) \\
 &= \overline{AB} + \overline{C}(\overline{A} + \overline{B}) \\
 &= MAJ(A, B, C)
 \end{aligned}$$

PGK

TABLE 11.1 Truth table for half adder

A	B	C _{out}	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

TABLE 11.2 Truth table for full adder

A	B	C	G	P	K	C _{out}	S
0	0	0	0	0	1	0	0
		1				0	1
0	1	0	0	1	0	0	1
		1				1	0
1	0	0	0	1	0	0	1
		1				1	0
1	1	0	1	0	0	1	0
		1				1	1

From the truth table, the half adder logic is

$$S = A \oplus B$$

$$C_{out} = A \cdot B$$

$$S = \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + \overline{A}BC + ABC$$

$$= (A \oplus B) \oplus C = P \oplus C$$

$$C_{out} = AB + AC + BC$$

$$= AB + C(A + B)$$

$$= \overline{AB} + \overline{C}(\overline{A} + \overline{B})$$

$$= \text{MAJ}(A, B, C)$$

- For a full adder, define what happens to carries (in terms of A and B)
- Generate: $C_{out} = 1$ independent of C
- $G = A \cdot B$
- Propagate: $C_{out} = C$
- $P = A \oplus B$
- Kill: $C_{out} = 0$ independent of C
- $K = \sim A \cdot \sim B$

Full Adder Design I (Static CMOS)

- ❑ Without any optimization requires 32 Transistors for Sum and 14 Transistors for Carry respectively
- ❑ Inverting all inputs to a full adder results in inverted values for all outputs

$$S = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$

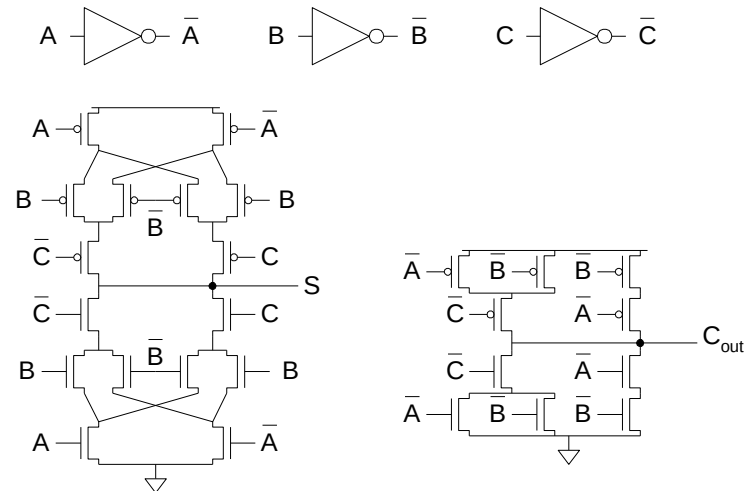
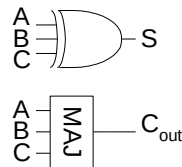
$$= (A \oplus B) \oplus C = P \oplus C$$

$$C_{out} = AB + AC + BC$$

$$= AB + C(A + B)$$

$$= \overline{AB + C}(\overline{A + B})$$

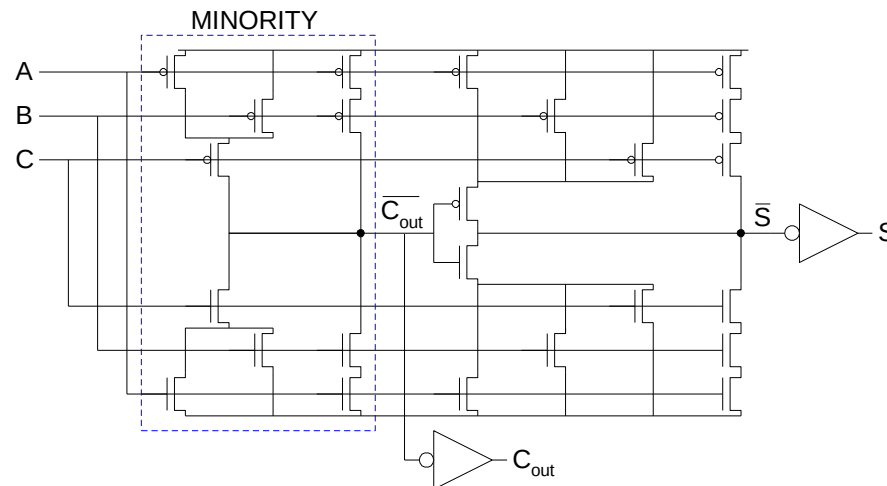
$$= \text{MAJ}(A, B, C)$$



- ❑ 32 Transistors

Full Adder Design II (Mirror Adder)

- ❑ Good to share some logic between sum & carry subcircuits that does not slow down carry generation
- ❑ Factor S in terms of C_{out}
$$S = ABC + (A + B + C)(\sim C_{out})$$
- ❑ Critical path is usually C to C_{out} in ripple adder



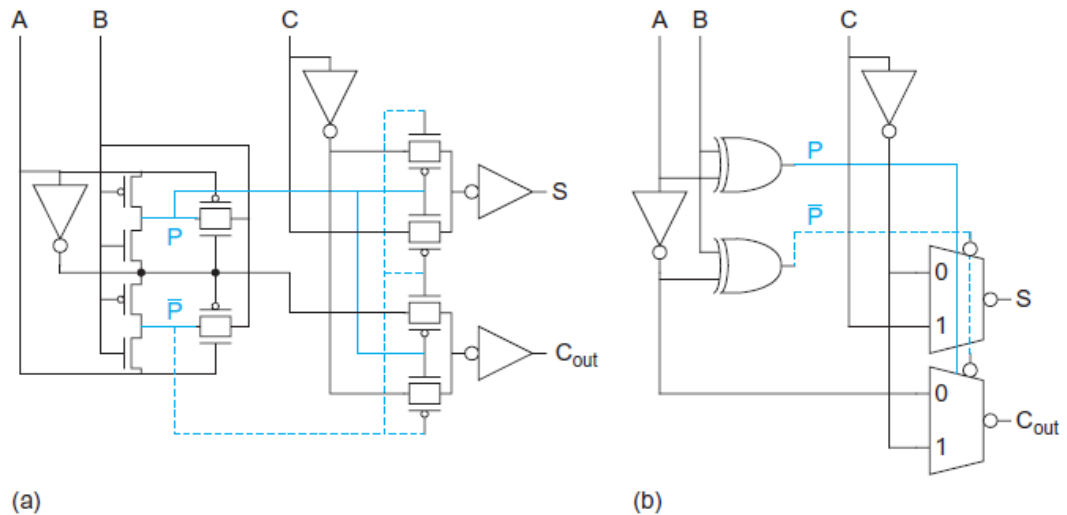
- ❑ 28 Transistors

Full Adder Design III

(Transmission Gate Full Adder)

- ❑ Parsing the transmission gate structures into MUX and an “invertible inverter” XOR structure

A	B	C	C _{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



- ❑ Note that the multiplexer choosing S is configured to compute $P \oplus C$
- ❑ 24 Transistors

Full Adder Design III

(Transmission Gate Full Adder)

A	B	C	C _{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

XOR/XNOR Circuit Forms

$$\square Y = A'B + AB' = AB + A'B'$$

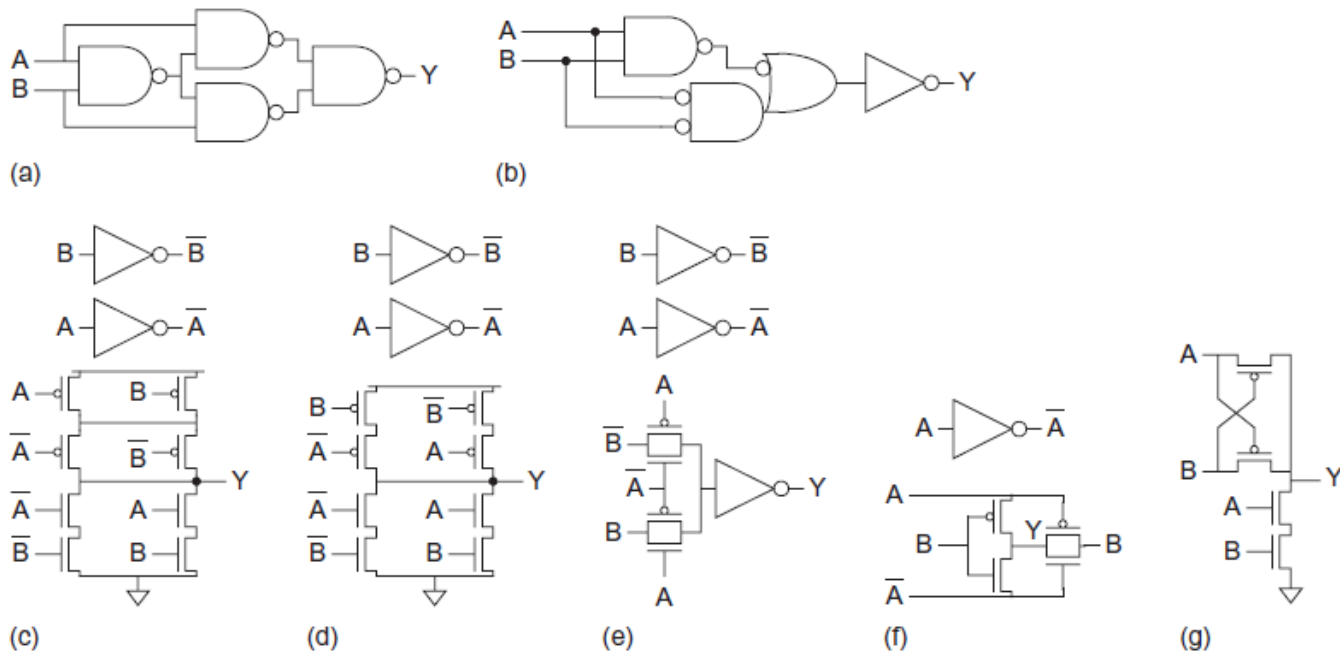
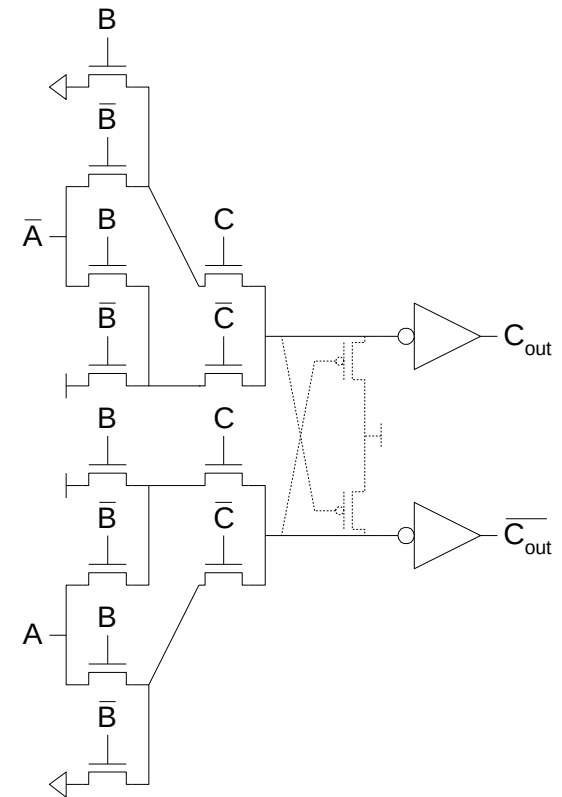
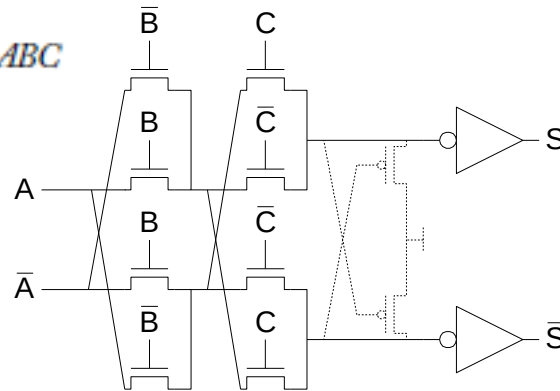


FIGURE 11.59 Static 2-input XOR designs

Full Adder Design IV (CPL)

- ❑ Complementary Pass Transistor Logic (CPL)
 - Slightly faster, but more area

$$\begin{aligned}
 S &= \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC \\
 &= (A \oplus B) \oplus C = P \oplus C \\
 C_{out} &= AB + AC + BC \\
 &= AB + C(A + B) \\
 &= \overline{\overline{AB} + \overline{C}(\overline{A} + \overline{B})} \\
 &= \text{MAJ}(A, B, C)
 \end{aligned}$$



Full Adder Design V (Dual-rail domino)

❑ Dual-rail domino

- Very fast, but large and power hungry
- Used in very fast multipliers

$$S = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$

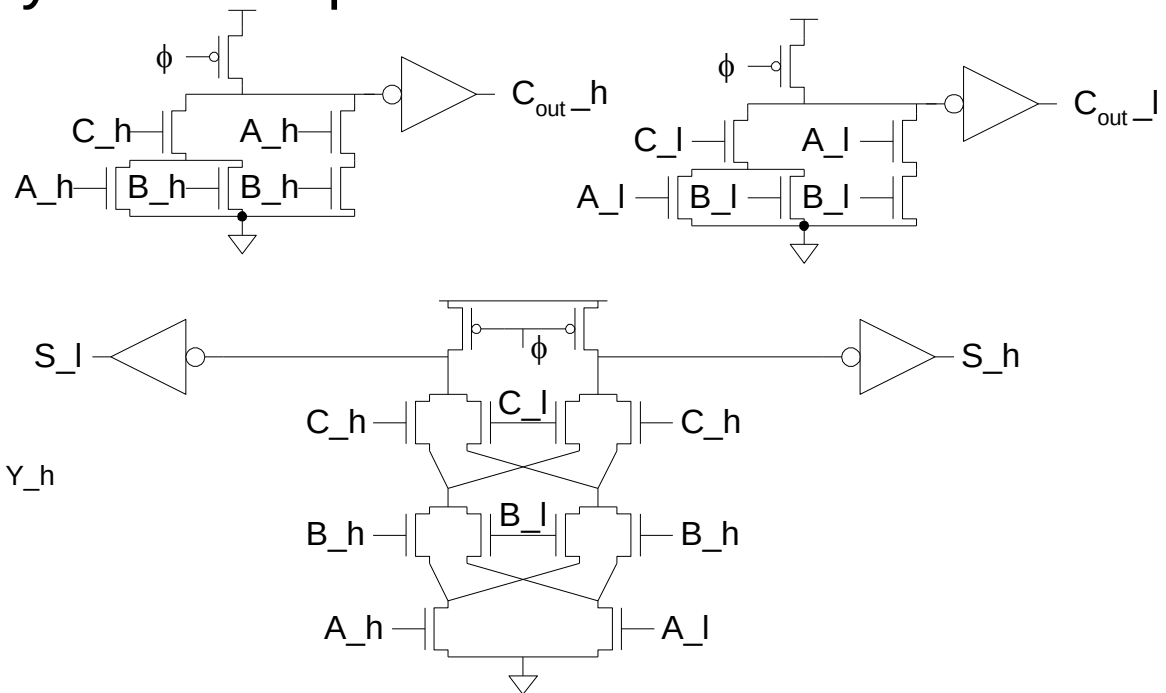
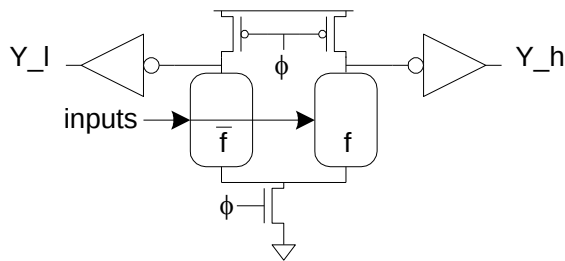
$$= (A \oplus B) \oplus C = P \oplus C$$

$$C_{out} = AB + AC + BC$$

$$= AB + C(A + B)$$

$$= \overline{\overline{AB} + \overline{C}(\overline{A} + \overline{B})}$$

$$= \text{MAJ}(A, B, C)$$



Home Work

- ❑ Implement the Difference output of Full Subtractor using 2x1 Mux at transistor level.
- ❑ Implement a single circuit for 4-bit Adder/Subtractor with control input using minimum resources at transistor level.
- ❑ Implement the following combinational circuit at transistor level with minimum number of resources:
 - Natural Binary Code to Gray code
 - 2-bit comparator